Construction Project Database Normalization

Group 3

Github: https://github.com/angelkibui/db-normalization-group-3
Activity Sheet:
https://docs.google.com/spreadsheets/d/122ulRz01Zpmo2_02bZiz66HJ6LJTY99kuaN4IVNORbM/edit?usp=sharing

We organised the construction data into smaller tables so each fact has a clear place. This document explains the changes from 1NF to 4NF, including the BCNF check, and the keys that connect the tables.

What we found in the raw data

The raw file has 15 rows and 26 columns. It describes 7 projects, 4 clients, 3 supervisors, 5 workers and 6 suppliers, along with materials and equipment.

Project details repeat for each worker or supplier listed on that project. Client contacts and supervisor phone numbers also repeat across projects. This can cause mistakes when information changes because someone might update one copy and miss another.

Projects can have several workers and suppliers, and a worker or supplier can appear on several projects. Each project has one listed client and supervisor. A client or supervisor can be linked to several projects.

Seven columns contain lists separated by a pipe character:

- WorkerSkills
- WorkerCertifications
- SupplierPhones
- MaterialSupplied
- MaterialUnitCost
- EquipmentUsed
- EquipmentRentalCost

A few terms used below

- A **primary key** identifies one row in a table.
- A **composite key** uses more than one column to identify a row.
- A **foreign key** connects a row to a key in another table.

- A **dependency** means that one value determines another. For example, under our chosen rules, a client's name identifies that client's phone number.

1. First Normal Form – 1NF

1NF means each cell holds one value and each row can be identified.

We kept the original single-value columns in **`Construction_1NF_Base.csv`** and gave each source row a **`SourceRowID`** from 1 to 15. We then split the lists into five child tables.

File	Primary key
WorkerSkills_1NF.csv	SourceRowID and Skill
WorkerCertifications_1NF.csv	SourceRowID and Certification
SupplierPhones_1NF.csv	SourceRowID and SupplierPhone
ProjectMaterials_1NF.csv	SourceRowID and Material
ProjectEquipment_1NF.csv	SourceRowID and Equipment

`SourceRowID` is the base table's primary key and a foreign key in every child table. It allows each child row to be connected to its original source record.

Both new rows and new tables were created. We kept each material with the cost in the same list position and did the same for equipment and rental costs.

For example, a source row listing Concrete and Steel with costs of 120 and 300 becomes one row for Concrete at 120 and another for Steel at 300.

The values are now atomic, meaning each cell contains one value. Repeated descriptive details remain for the next stages.

2. Second Normal Form – 2NF

2NF removes partial dependencies. This means a descriptive value must depend on the whole key, rather than only part of a composite key.

For example, the 1NF skill table uses `SourceRowID` and `Skill` to identify a row. However, the worker's name and phone number depend on `SourceRowID` alone. Adding another skill should not require another copy of the worker's phone number.

We separated the project, worker and supplier details:

- Project details depend on ProjectID.
- A worker's phone number and hourly rate depend on WorkerName.
- A supplier's city depends on SupplierName.

These facts were moved into their own entity tables. Separate relationship tables record which workers and suppliers belong to each project.

File	Primary key
Projects_2NF.csv	ProjectID
Workers_2NF.csv	WorkerName
Suppliers_2NF.csv	SupplierName
ProjectWorkers_2NF.csv	ProjectID and WorkerName
ProjectSuppliers_2NF.csv	ProjectID and SupplierName
WorkerSkills_2NF.csv	WorkerName and Skill
WorkerCertifications_2NF.csv	WorkerName and Certification
SupplierPhones_2NF.csv	SupplierName and SupplierPhone
ProjectMaterials_2NF.csv	ProjectID, SupplierName and Material
ProjectEquipment_2NF.csv	ProjectID and Equipment

How the tables connect

ProjectWorkers connects Projects to Workers. ProjectSuppliers connects Projects to Suppliers.

WorkerSkills and WorkerCertifications reference Workers. SupplierPhones references Suppliers. ProjectMaterials references the matching project and supplier combination in ProjectSuppliers. ProjectEquipment references Projects.

These are logical foreign keys. CSV files store the data but do not enforce database constraints themselves.

MaterialUnitCost stays with the project, supplier and material combination. EquipmentRentalCost stays with the project and equipment combination because these costs describe those relationships.

Client and supervisor details still remain in Projects. Their dependencies are addressed in 3NF.

Assumptions carried forward

We use names as keys because separate IDs are not provided. We assume each name identifies one entity.

We treat skills and certifications as facts about a worker, combining the distinct values reported across their source rows. For example, Mike Ross has Carpentry, Framing and Concrete in the resulting skill table.

This follows the existing 2NF approach. It does not record which skill was listed on which project or connect a particular worker to a supplier. Those original row associations remain available in 1NF through SourceRowID. If those associations have a separate business meaning, they would need their own relationship tables.

Under these assumptions, the descriptive columns depend on their whole keys. The remaining client and supervisor dependencies are handled in the next stage.

3. Third Normal Form - 3NF

A table must already be in 2NF before it can meet 3NF.

This stage removes transitive dependencies. In simple terms, this means moving details that depend on another non-key column into their own table.

In **`Projects_2NF.csv`**, the project identifies its client, while the client's name identifies the client's contact details.

Dependencies we found

Each project has a client. The client's phone number, email address and city depend on the client's name, rather than on the project itself.

Each project also has a supervisor. The supervisor's phone number depends on the supervisor's name, rather than on the project itself.

ClientName and SupervisorName are not keys in Projects because the same client or supervisor can appear on several projects. Their contact details should be stored once.

Changes made

We created two tables and updated Projects.

File	Columns

Projects_3NF.csv	ProjectID, ProjectName, ProjectType, StartDate, EndDate, SiteAddress, SiteCity, SiteState, ClientName and SupervisorName
Clients_3NF.csv	ClientName, ClientPhone, ClientEmail and ClientCity
Supervisors_3NF.csv	SupervisorName and SupervisorPhone

The primary keys are:

- ProjectID for Projects.
- ClientName for Clients.
- SupervisorName for Supervisors.

Projects keeps ClientName as a foreign key to Clients and SupervisorName as a foreign key to Supervisors.

For example, P001 and P005 both refer to Metro Corp. Its phone number, email address and city now appear in one client row. If its phone number changes, only that row needs updating.

The other nine 2NF tables are carried forward with the **`_3NF.csv`** suffix. Their columns, keys and relationships remain the same. The 3NF folder therefore contains 12 tables.

How we preserved the relationships

Each project points to exactly one matching client row and one matching supervisor row.

Joining those rows back to Projects restores every column and value in the original 2NF project table without adding or losing project rows. This is called a lossless decomposition.

Why the design meets 3NF

Client contact details now depend directly on the key of Clients. Supervisor phone numbers depend directly on the key of Supervisors.

Project descriptions and dates stay with ProjectID. Worker and supplier details already have their own tables.

Under the stated dependency rules, no non-key column determines another non-key column within these tables.

We keep site address, city and state together as project details. A city name alone is not a reliable identifier for a state, and a street address alone is not a reliable location identifier. We do not assume either rule from this small dataset.

4. Boyce-Codd Normal Form - BCNF

BCNF is a stricter check than 3NF.

When a column or combination of columns determines values in other columns, it must also identify a whole row. The determining columns are called a **determinant**.

Columns that identify a row form a **superkey**. If none of those columns can be removed without losing that ability, they form a **candidate key**.

Dependencies and keys checked

Table	What determines the details and identifies a row
Projects	ProjectID determines the project details and is the candidate key.
Clients	ClientName determines the client's phone number, email and city. It is the candidate key.
Supervisors	SupervisorName determines the supervisor's phone number and is the candidate key.
Workers	WorkerName determines the worker's phone number and hourly rate. It is the candidate key.
Suppliers	SupplierName determines the supplier's city and is the candidate key.
ProjectMaterials	ProjectID, SupplierName and Material together determine the unit cost and form the candidate key.
ProjectEquipment	ProjectID and Equipment together determine the rental cost and form the candidate key.
ProjectWorkers	ProjectID and WorkerName together form the candidate key.
ProjectSuppliers	ProjectID and SupplierName together form the candidate key.
WorkerSkills	WorkerName and Skill together form the candidate key.
WorkerCertifications	WorkerName and Certification together form the candidate key.
SupplierPhones	SupplierName and SupplierPhone together form the candidate key.

The last five tables contain only their key columns. They record relationships without adding descriptive details that belong elsewhere.

Why costs remain with their relationships

An equipment name does not determine one rental cost. For example, Crane costs 5000 on P001 and 5500 on P002.

A supplier and material do not determine one unit cost across all projects either. SteelWorks Inc supplies Steel Beams at 500 on P001 and 520 on P007. The project must therefore remain part of these cost records.

We also allow different suppliers to quote different prices for the same material on a project. The two P004 Glass Panels entries happen to have the same price. That does not establish a general rule that all suppliers must charge the same amount.

We do not assume that phone numbers, email addresses or project names must be unique.

Result of the BCNF check

No table violates BCNF under these rules, so no further decomposition is needed.

The BCNF folder contains the same 12 tables as 3NF, renamed with the **`_BCNF.csv`** suffix. Their candidate keys remain the same.

A small dataset can show that a dependency is false, but it cannot prove that a rule will hold for every future record. The design therefore relies on the business assumptions explained above.

5. Fourth Normal Form – 4NF

A table must meet BCNF before it can meet 4NF.

This stage checks for independent lists of facts about the same entity. A problem arises when the entity has several values in each list and storing them together requires repeated combinations.

Skills and certifications

One worker can have several skills and several certifications.

We treat these as independent lists because a skill entry is not paired with a particular certification. These are multivalued dependencies: the worker is associated with a set of skills and a separate set of certifications.

Mike Ross has three recorded skills and two certifications. A table combining both independent lists would need six rows to include every skill with every certification.

Separate tables need three skill rows and two certification rows. A new certification can then be added without repeating every skill.

File	Primary key	Rows
WorkerSkills_4NF.csv	WorkerName and Skill	15
WorkerCertifications_4NF.csv	WorkerName and Certification	12
SupplierPhones_4NF.csv	SupplierName and SupplierPhone	8

Supplier phone numbers

A supplier's phone numbers are independent of the materials it supplies on projects. Combining those facts would repeat each material for each phone number.

SupplierPhones stores the phone list. ProjectMaterials stores each project, supplier, material and cost fact.

For example, BuildPro Supplies has two phone numbers. Those phone numbers do not need to be repeated beside both Concrete and Steel on P001.

Facts that must stay together

Materials and their costs are paired facts. Equipment items and their rental costs are also paired facts.

Splitting either pair into independent lists would lose which cost belongs to which item.

The source contains no equipment training or inspection lists, so no tables are needed for those examples mentioned in the activity.

What changed at this stage

The earlier work had already separated the independent lists, so no new split was needed.

WorkerSkills records one worker and one skill per row. WorkerCertifications records one worker and one certification per row. SupplierPhones records one supplier and one phone number per row.

None of these tables mixes two independent lists.

The 4NF folder contains the same 12 tables as BCNF, renamed with the `_4NF.csv` suffix.

Under our assumptions, the tables meet BCNF and have no remaining independent lists that need separating. Each descriptive fact or relationship can be recorded without repeating unrelated list entries.

Checks on the completed files

We checked that:

- Each cell contains one value.
- Every declared primary key is filled and unique.
- Every foreign key has a matching parent record.
- Materials remain paired with their correct unit costs.
- Equipment items remain paired with their correct rental costs.
- Joining Clients and Supervisors back to Projects restores all seven 2NF project rows exactly.
- The original 1NF and 2NF files remain unchanged.

Each of the final three stages contains 7 projects, 4 clients, 3 supervisors, 5 workers, 6 suppliers, 15 project-worker links, 10 project-supplier links, 16 material records, 13 equipment records, 15 worker skills, 12 worker certifications and 8 supplier phone numbers.